Lab 03: Azure Firewall

Student lab manual

Lab scenario

You have been asked to install Azure Firewall. This will help your organization control inbound and outbound network access which is an important part of an overall network security plan. Specifically, you would like to create and test the following infrastructure components:

- A virtual network with a workload subnet and a jump host subnet.
- A virtual machine in each subnet.
- A custom route that ensures all outbound workload traffic from the workload subnet must use the firewall.
- Firewall Application rules that only allow outbound traffic to www.bing.com.
- Firewall Network rules that allow external DNS server lookups.

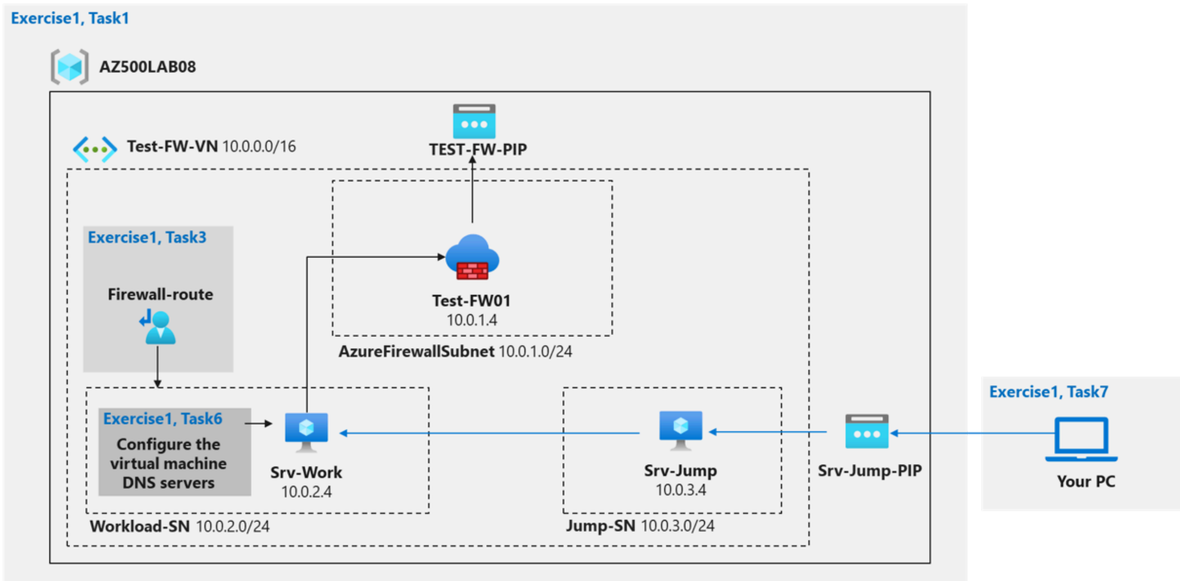
For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

Lab objectives

In this lab, you will complete the following exercise:

- Exercise 1: Deploy and test an Azure Firewall

Azure Firewall diagram



Instructions

Lab files:

- **\Allfiles\Labs\08\template.json**

Exercise 1: Deploy and test an Azure Firewall

Estimated timing: 40 minutes

For all the resources in this lab, we are using the **East (US)** region. Verify with your instructor this is the region to use for your class.

In this exercise, you will complete the following tasks:

- Task 1: Use a template to deploy the lab environment.
- Task 2: Deploy an Azure firewall.
- Task 3: Create a default route.
- Task 4: Configure an application rule.

- Task 5: Configure a network rule.
- Task 6: Configure DNS servers.
- Task 7: Test the firewall.

Task 1: Use a template to deploy the lab environment.

In this task, you will review and deploy the lab environment.

In this task, you will create a virtual machine by using an ARM template. This virtual machine will be used in the last exercise for this lab.

1. Sign-in to the Azure portal `https://portal.azure.com/`.
   Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is LabUser-63145501@LODSPRODMSLEARNMCA.onmicrosoft.com
2. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Deploy a custom template** and press the **Enter** key.
3. On the **Custom deployment** blade, click the **Build your own template in the editor** option.
4. On the **Edit template** blade, click **Load file**, locate the **\Allfiles\Labs\08\template.json** file and click **Open**.
   Review the content of the template and note that it deploys an Azure VM hosting Windows Server 2016 Datacenter.
5. On the **Edit template** blade, click **Save**.

6. On the **Custom deployment** blade, ensure that the following settings are configured (leave any others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	Use existing Resource Group **AZ500LAB08**
Location	**(US) East US**
adminPassword	A secure password of your own choosing for the virtual machines. Remember the password. You will need it later to connect to the VMs.

7. To identify Azure regions where you can provision Azure VMs, refer to https://azure.microsoft.com/en-us/regions/offers/
8. Click **Review + create**, and then click **Create**.
   Wait for the deployment to complete. This should take about 2 minutes.

Task 2: Deploy the Azure firewall

If the firewall deployment fails with an error message stating, 'internalservererror,' delete all deployed resources in the Azure portal via the All Resources blade, and then edit the template.json file from task 1 to replace all mentions of 'eastus' with a different region such as 'centralus.' After making the update, re-deploy the template and Azure firewall using the newly specified region for all deployments. Please note: When deploying the template, the Azure portal will still show eastus as the region. This can be ignored as long as all mentions of eastus in the template.json file have been replaced with the new region.

In this task you will deploy the Azure firewall into the virtual network.

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Firewalls** and press the **Enter** key.
2. On the **Firewalls** blade, click **+ Create**.
3. On the **Basics** tab of the **Create a firewall** blade, specify the following settings:

Setting	Value
Resource group	AZ500LAB08
Name	Test-FW01
Region	**(US) East US**
Firewall SKU	Standard
Firewall management	Use Firewall rules (classic) to manage this firewall
Choose a virtual network	click the **Use existing** option and, in the drop-down list, select **Test-FW-VN**
Firewall Management NIC	To disable this feature, **deselect** the **Enable Firewall Management NIC** option.
Public IP address	click **Add new** and type the name TEST-FW-PIP and click **OK**

4. Click **Review + create** and then click **Create**.
   Wait for the deployment to complete. This should take about 5 minutes.
5. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.

6. On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB08** entry.
   On the **AZ500LAB08** resource group blade, review the list of resources. You can sort by **Type**.
7. In the list of resources, click the entry representing the **Test-FW01** firewall.
8. On the **Test-FW01** blade, identify the **Private IP** address that was assigned to the firewall.
   You will need this information in the next task.

Task 3: Create a default route

In this task, you will create a default route for the **Workload-SN** subnet. This route will configure outbound traffic through the firewall.

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Route tables** and press the **Enter** key.
2. On the **Route tables** blade, click **+ Create**.
3. On the **Create route table** blade, specify the following settings:

Setting	Value
Resource group	AZ500LAB08
Region	**East US**
Name	Firewall-route

4. Click **Review + create**, then click **Create**, and wait for the provisioning to complete.
5. On the **Route tables** blade, click **Refresh**, and, in the list of route tables, click the **Firewall-route** entry.
6. On the **Firewall-route** blade, in the **Settings** section, click **Subnets** and then, on the **Firewall-route | Subnets** blade, click **+ Associate**.
7. On the **Associate subnet** blade, specify the following settings:

Setting	Value
Virtual network	Test-FW-VN
Subnet	Workload-SN

8. Ensure the **Workload-SN** subnet is selected for this route, otherwise the firewall won't work correctly.
9. Click **OK** to associate the firewall to the virtual network subnet.

10. Back on the **Firewall-route** blade, in the **Settings** section, click **Routes** and then click **+ Add**.
11. On the **Add route** blade, specify the following settings:

Setting	Value
Route name	FW-DG
Destination Type	IP Address
Destination IP addresses/CIDR ranges	**0.0.0.0/0**
Next hop type	Virtual appliance
Next hop address	the private IP address of the firewall that you identified in the previous task

12. Azure Firewall is actually a managed service, but virtual appliance works in this situation.
13. Click **Add** to add the route.

Task 4: Configure an application rule

In this task you will create an application rule that allows outbound access to `www.bing.com`.

1. In the Azure portal, navigate back to the **Test-FW01** firewall.
2. On the **Test-FW01** blade, in the **Settings** section, click **Rules (classic)**.
3. On the **Test-FW01 | Rules (classic)** blade, click the **Application rule collection** tab, and then click **+ Add application rule collection**.
4. On the **Add application rule collection** blade, specify the following settings (leave others with their default values):

Setting	Value
Name	App-Coll01
Priority	200
Action	Allow

5. On the **Add application rule collection** blade, create a new entry in the **Target FQDNs** section with the following settings (leave others with their default values):

Setting	Value
name	AllowGH

Source type	IP Address
Source	10.0.2.0/24
Protocol port	http:80, https:443
Target FQDNS	www.bing.com

6. Click **Add** to add the Target FQDNs-based application rule.
   Azure Firewall includes a built-in rule collection for infrastructure FQDNs that are allowed by default. These FQDNs are specific for the platform and can't be used for other purposes.

Task 5: Configure a network rule

In this task, you will create a network rule that allows outbound access to two IP addresses on port 53 (DNS).

1. In the Azure portal, navigate back to the **Test-FW01 | Rules (classic)** blade.
2. On the **Test-FW01 | Rules (classic)** blade, click the **Network rule collection** tab and then click **+ Add network rule collection**.

3. On the **Add network rule collection** blade, specify the following settings (leave others with their default values):

Setting	Value
Name	Net-Coll01
Priority	200
Action	Allow

4. On the **Add network rule collection** blade, create a new entry in the **IP Addresses** section with the following settings (leave others with their default values):

Setting	Value
Name	AllowDNS
Protocol	UDP
Source type	IP address
Source Addresses	10.0.2.0/24
Destination type	IP address

Destination Address	209.244.0.3,209.244.0.4
Destination Ports	53

5. Click **Add** to add the network rule.
   The destination addresses used in this case are known public DNS servers.

Task 6: Configure the virtual machine DNS servers

In this task, you will configure the primary and secondary DNS addresses for the virtual machine. This is not a firewall requirement.

1. In the Azure portal, navigate back to the **AZ500LAB08** resource group.
2. On the **AZ500LAB08** blade, in the list of resources, click the **Srv-Work** virtual machine.
3. On the **Srv-Work** blade, click **Networking**.
4. On the **Srv-Work | Networking Settings** blade, click the link next to the **Network interface** entry.
5. On the network interface blade, in the **Settings** section, click **DNS servers**, select the **Custom** option, add the two DNS servers referenced in the network rule: **209.244.0.3** and **209.244.0.4**, and click **Save** to save the change.
6. Return to the **Srv-Work** virtual machine page.
   Wait for the update to complete.
   Updating the DNS servers for a network interface will automatically restart the virtual machine to which that interface is attached, and if applicable, any other virtual machines in the same availability set.

Task 7: Test the firewall

In this task, you will test the firewall to confirm that it works as expected.

1. In the Azure portal, navigate back to the **AZ500LAB08** resource group.
2. On the **AZ500LAB08** blade, in the list of resources, click the **Srv-Jump** virtual machine.
3. On the **Srv-Jump** blade, click **Connect** and, in the drop down menu, click **Connect**.

4. Download the RDP file and use it to connect to the **Srv-Jump** Azure VM via Remote Desktop. When prompted to authenticate, provide the following credentials:

Setting	Value
User name	localadmin

Password	The secure password you chose during deployment of the custom template in task 1 step 6.

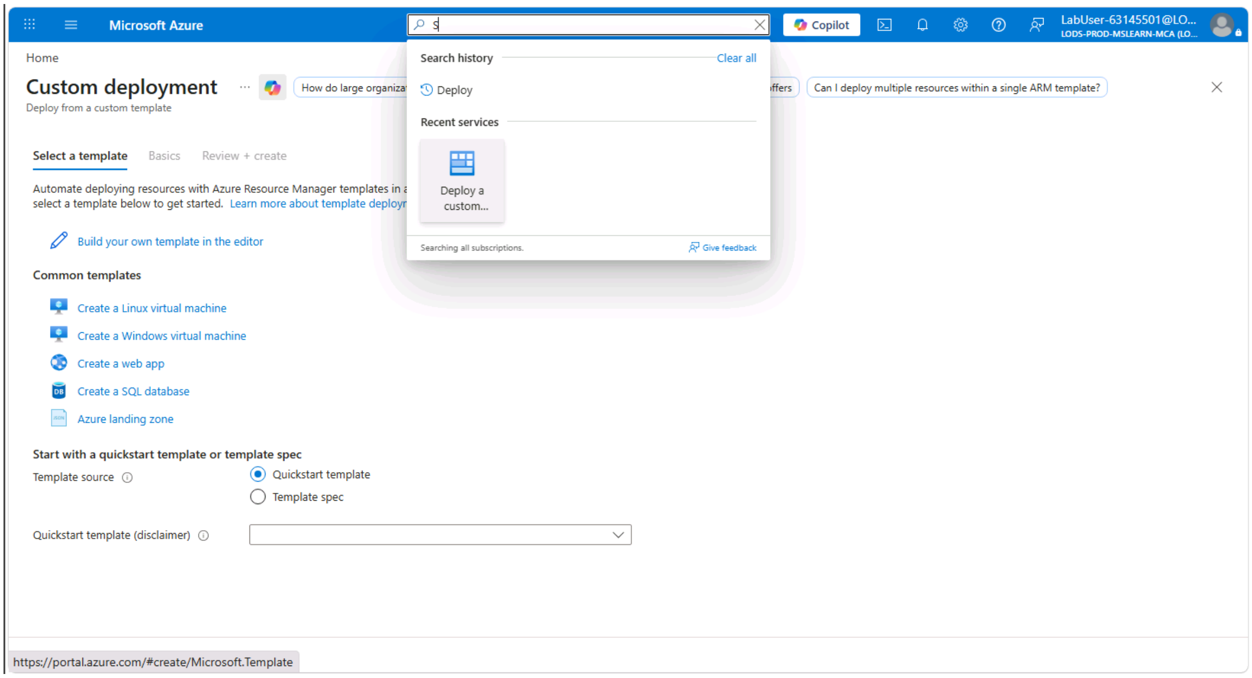
5. The following steps are performed in the Remote Desktop session to the **Srv-Jump** Azure VM.
   You will connect to the **Srv-Work** virtual machine. This is being done so we can test the ability to access the bing.com website.
6. Within the Remote Desktop session to **Srv-Jump**, right-click **Start**, in the right-click menu, click **Run**, and, from the **Run** dialog box, run the following to connect to **Srv-Work**.
7. `mstsc /v:Srv-Work`
8. When prompted to authenticate, provide the following credentials:

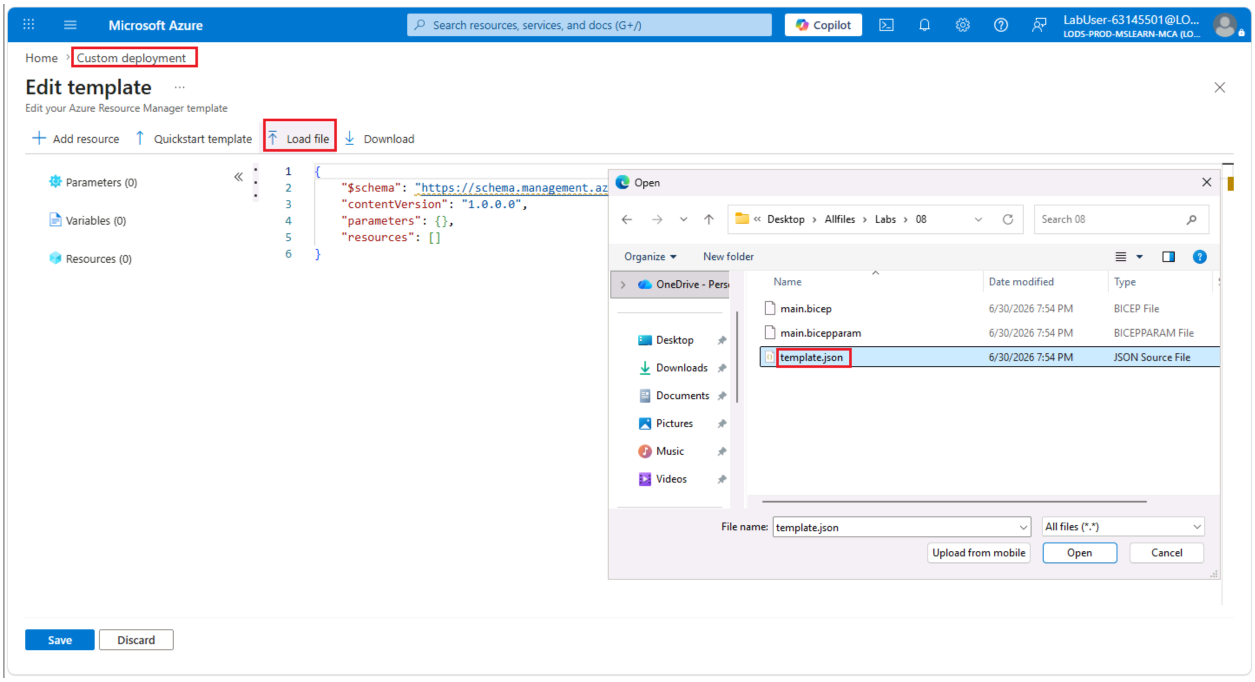
Setting	Value
User name	localadmin
Password	The secure password you chose during deployment of the custom template in task 1 step 6.

9. Wait for the Remote Desktop session to be established and the Server Manager interface to load.
10. Within the Remote Desktop session to **Srv-Work**, in **Server Manager**, click **Local Server** and then click **IE Enhanced Security Configuration**.
11. In the **Internet Explorer Enhanced Security Configuration** dialog box, set both options to **Off** and click **OK**.
12. Within the Remote Desktop session to **Srv-Work**, start Internet Explorer and browse to `https://www.bing.com`.
    The website should successfully display. The firewall allows you access.
13. Browse to `http://www.microsoft.com/`
    Within the browser page, you should receive a message with text resembling the following: `HTTP request from 10.0.2.4:xxxxx to microsoft.com:80. Action: Deny. No rule matched. Proceeding with default action.`
    This is expected, since the firewall blocks access to this website.
14. Terminate both Remote Desktop sessions.

Result: You have successfully configured and tested the Azure Firewall.

Screenshots




Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
LODS-PROD-MSLEARN-MCA (LO...
Home
Custom deployment
Edit template
Edit your Azure Resource Manager template
Add resource
Quickstart template
Load file
Download
Parameters (0)
Variables (0)
Resources (0)
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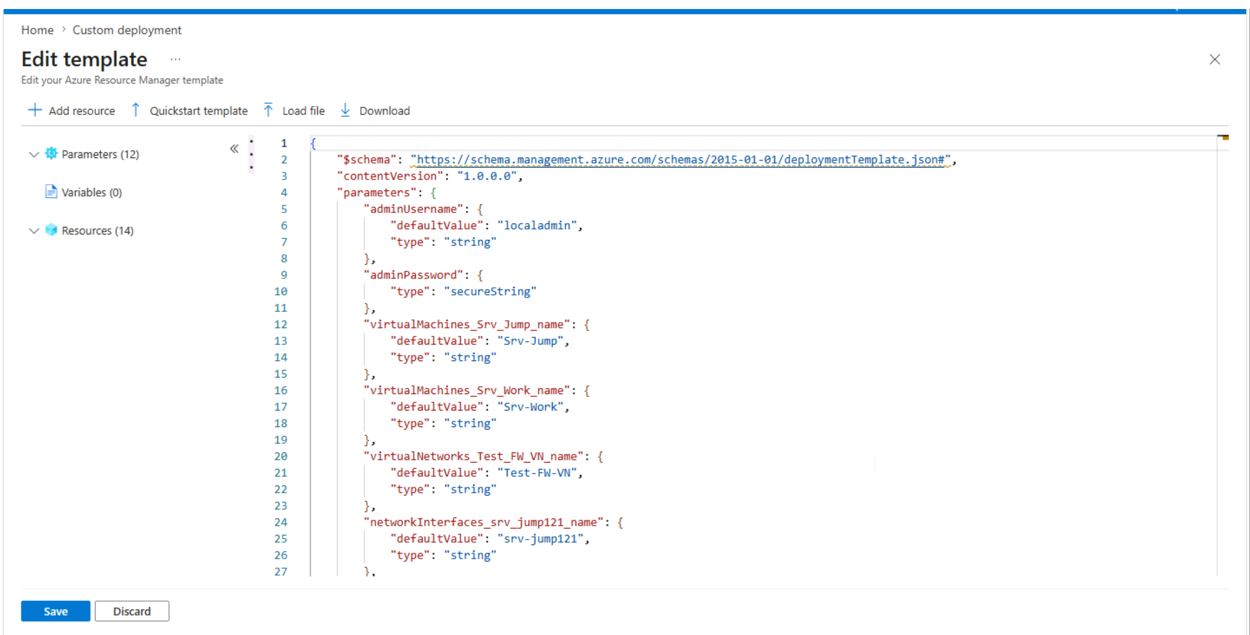
"parameters": {},
"resources": []
}
Open
Desktop › Allfiles › Labs › 08
Search 08
Organize
New folder
OneDrive - Pers
Desktop
Downloads
Documents
Pictures
Music
Videos
Name
Date modified
Type
main.bicep
6/30/2026 7:54 PM
BICEP File
main.bicepparam
6/30/2026 7:54 PM
BICEPPARAM File
template.json
6/30/2026 7:54 PM
JSON Source File
File name:
template.json
All files (*.*)
Upload from mobile
Open
Cancel
Save
Discard


Home
Custom deployment
Edit template
Edit your Azure Resource Manager template
Add resource
Quickstart template
Load file
Download
Parameters (12)
Variables (0)
Resources (14)
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"type": "string"
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"virtualMachines_Srv_Work_name": {
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"type": "string"
},
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"type": "string"
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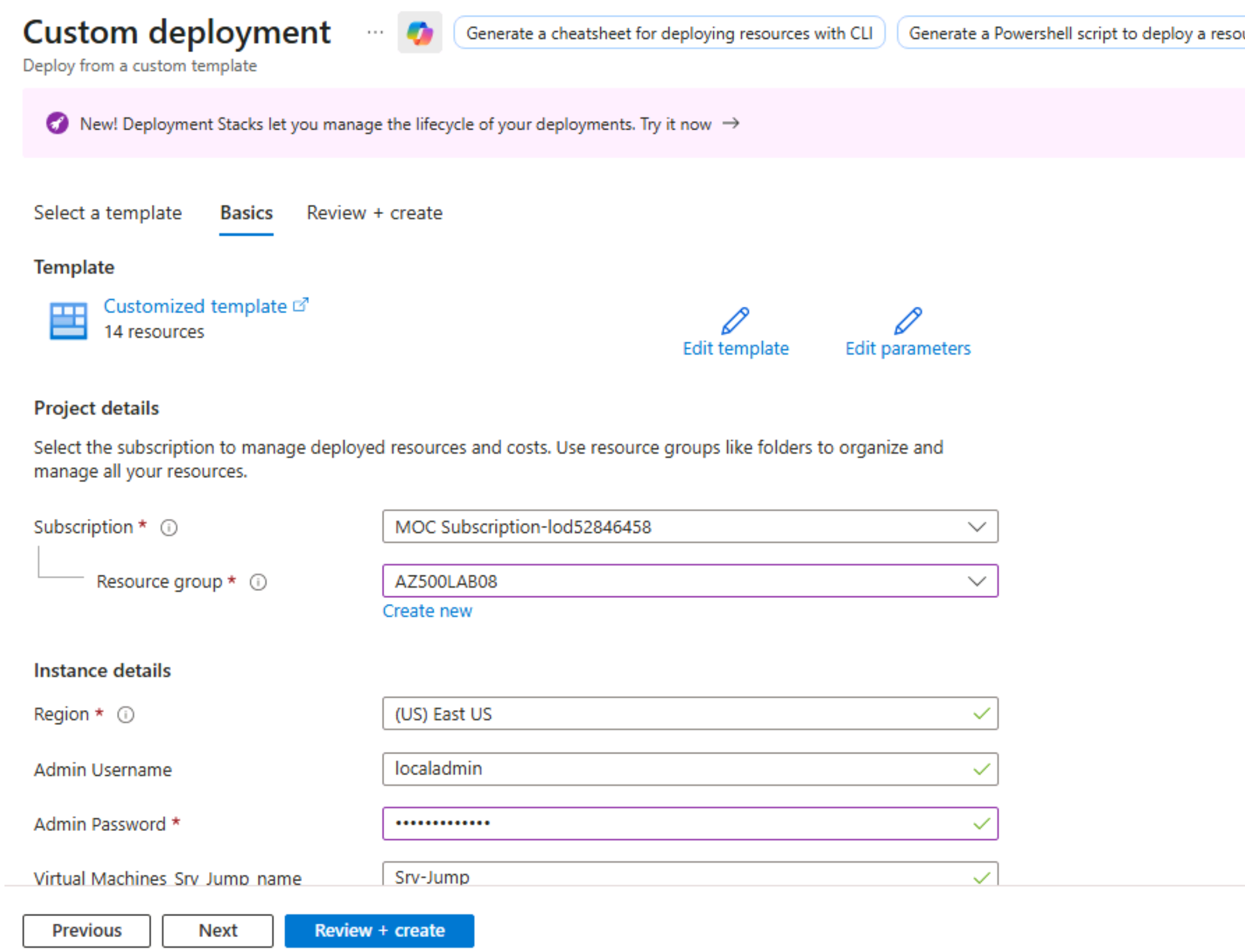
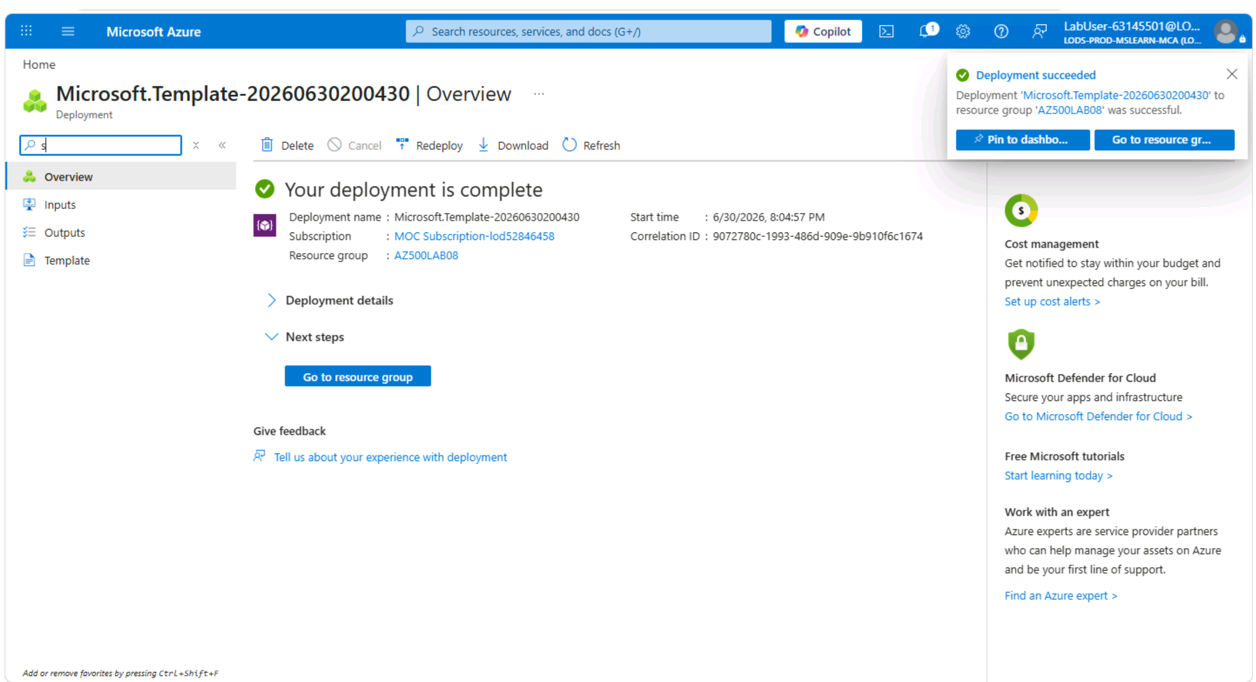
"type": "string"
},
Save
Discard


Custom deployment
Deploy from a custom template
Generate a cheatsheet for deploying resources with CLI
Generate a Powershell script to deploy a resou
New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →
Select a template
Basics
Review + create
Template
Customized template
14 resources
Edit template
Edit parameters
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
MOC Subscription-lod52846458
Resource group *
AZ500LAB08
Create new
Instance details
Region *
(US) East US
Admin Username
localadmin
Admin Password *
Virtual Machines_Srv_Jump_name
Srv-Jump
Previous
Next
Review + create


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
LODS-PROD-MSLEARN-MCA (LO...
Home
Microsoft.Template-20260630200430 | Overview
Deployment
Delete
Cancel
Redeploy
Download
Refresh
Overview
Inputs
Outputs
Template
Deployment succeeded
Deployment 'Microsoft.Template-20260630200430' to resource group 'AZ500LAB08' was successful.
Pin to dashbo...
Go to resource gr...
Your deployment is complete
Deployment name : Microsoft.Template-20260630200430
Subscription : MOC Subscription-lod52846458
Resource group : AZ500LAB08
Start time : 6/30/2026, 8:04:57 PM
Correlation ID : 9072780c-1993-486d-909e-9b910f6c1674
Deployment details
Next steps
Go to resource group
Give feedback
Tell us about your experience with deployment
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
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Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
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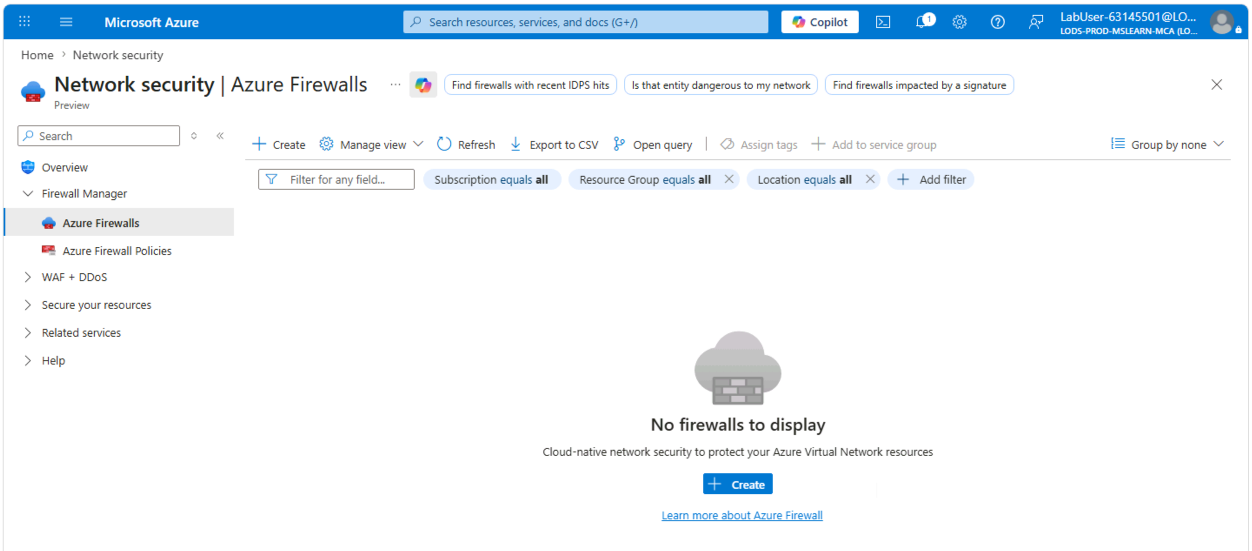
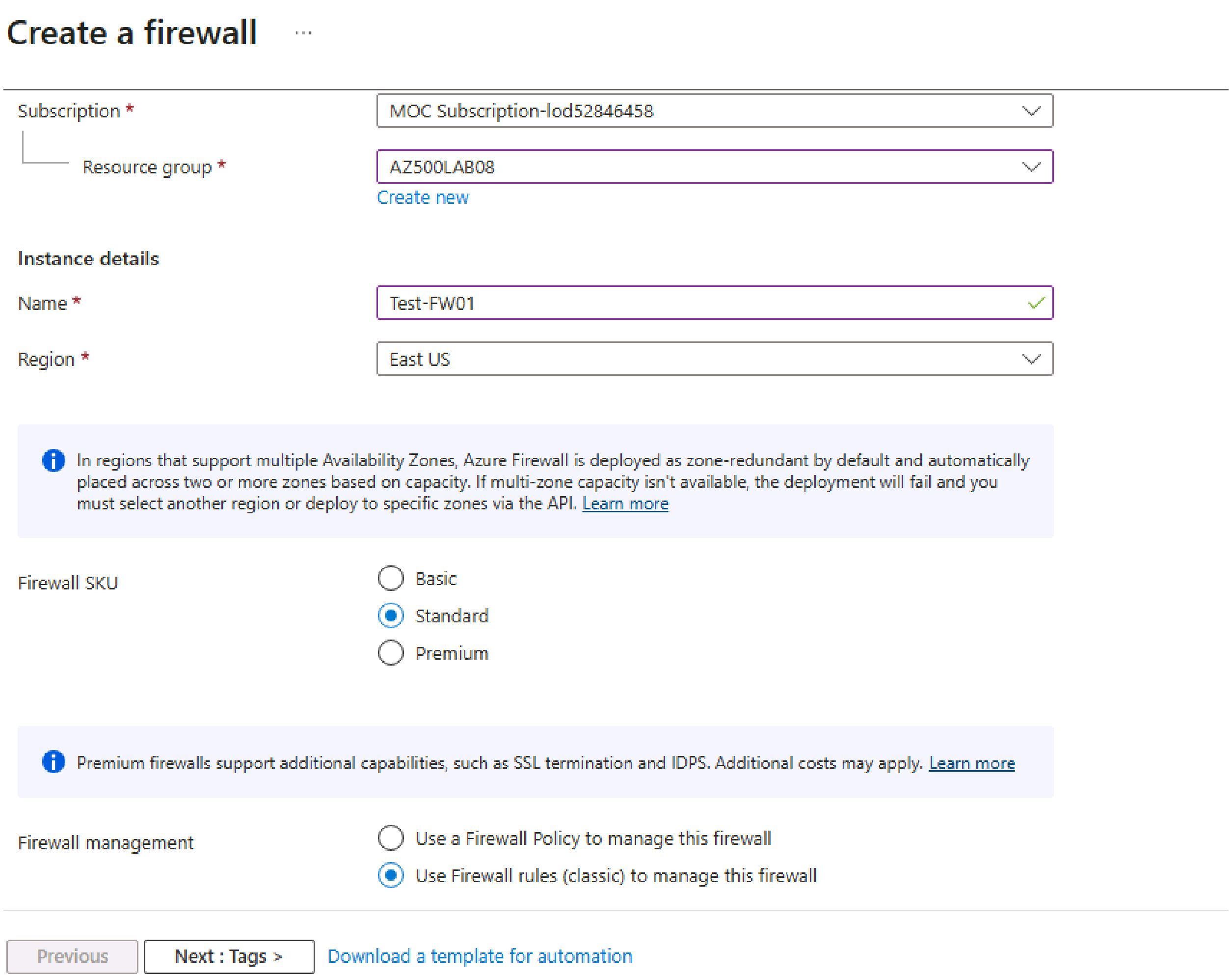

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
LODS-PROD-MSLEARN-MCA (LO...
Home > Network security
Network security | Azure Firewalls
Preview
Find firewalls with recent IDPS hits
Is that entity dangerous to my network
Find firewalls impacted by a signature
Search
Overview
Firewall Manager
Azure Firewalls
Azure Firewall Policies
WAF + DDoS
Secure your resources
Related services
Help
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
No firewalls to display
Cloud-native network security to protect your Azure Virtual Network resources
Create
Learn more about Azure Firewall


Create a firewall
Subscription *
MOC Subscription-lod52846458
Resource group *
AZ500LAB08
Create new
Instance details
Name *
Test-FW01
Region *
East US
In regions that support multiple Availability Zones, Azure Firewall is deployed as zone-redundant by default and automatically placed across two or more zones based on capacity. If multi-zone capacity isn't available, the deployment will fail and you must select another region or deploy to specific zones via the API. Learn more
Firewall SKU
Basic
Standard
Premium
Premium firewalls support additional capabilities, such as SSL termination and IDPS. Additional costs may apply. Learn more
Firewall management
Use a Firewall Policy to manage this firewall
Use Firewall rules (classic) to manage this firewall
Previous
Next : Tags >
Download a template for automation

Home > Network security | Azure Firewalls

Create a firewall ...

Standard

Premium

Premium firewalls support additional capabilities, such as SSL termination and IDPS. Additional costs may apply. Learn more

Firewall management
- Use a Firewall Policy to manage this firewall
- Use Firewall rules (classic) to manage this firewall

Choose a virtual network
- Create new
- Use existing

Virtual network: Test-FW-VN (AZ500LAB08)

Public IP address *: (New) TEST-FW-PIP

Add new

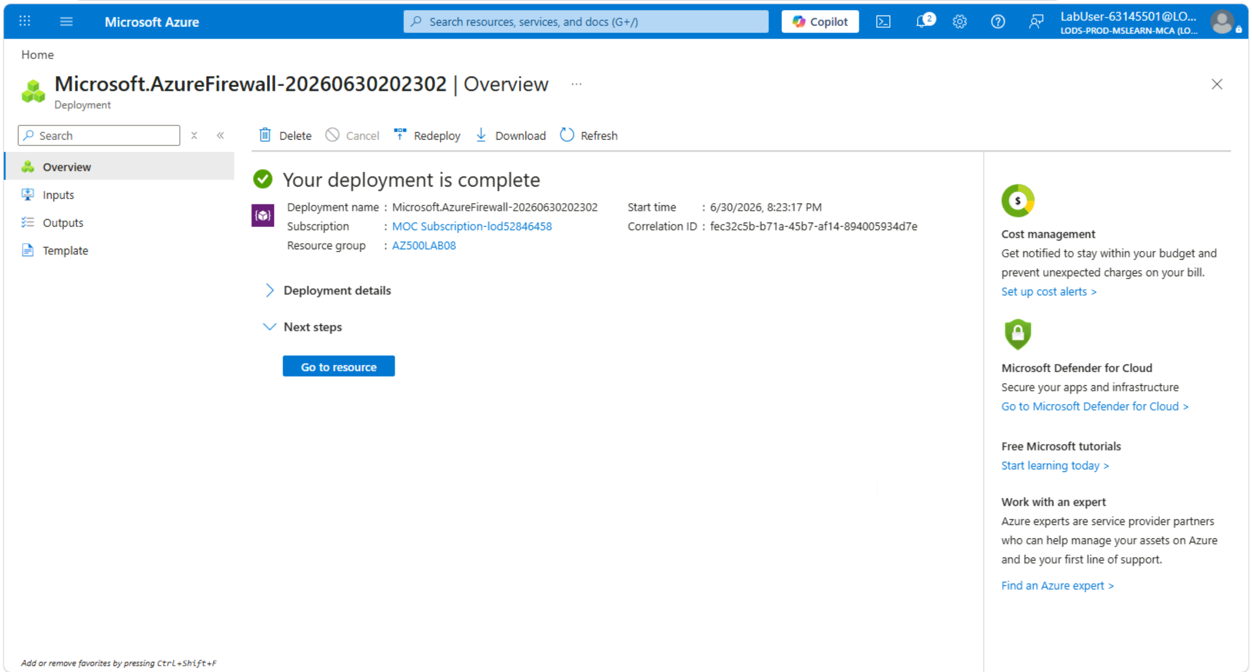
Firewall Management NIC

Firewall Management NIC separates Firewall management traffic from customer traffic. A dedicated subnet is required with its own associated public IP address that will be used exclusively by the Azure platform and can't be used for any other purpose.

Enable Firewall Management NIC

Critical Firewall features such as Forced Tunneling and Packet Capture require management NIC to be enabled.

Previous | Next : Tags > | Download a template for automation


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
LODS-PROD-MSLEARN-MCA (LO...
Home
Microsoft.AzureFirewall-20260630202302 | Overview
Deployment
Search
Delete
Cancel
Redeploy
Download
Refresh
Overview
Inputs
Outputs
Template
Your deployment is complete
Deployment name : Microsoft.AzureFirewall-20260630202302
Subscription : MOC Subscription-lod52846458
Resource group : AZ500LAB08
Start time : 6/30/2026, 8:23:17 PM
Correlation ID : fec32c5b-b71a-45b7-af14-894005934d7e
Deployment details
Next steps
Go to resource
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
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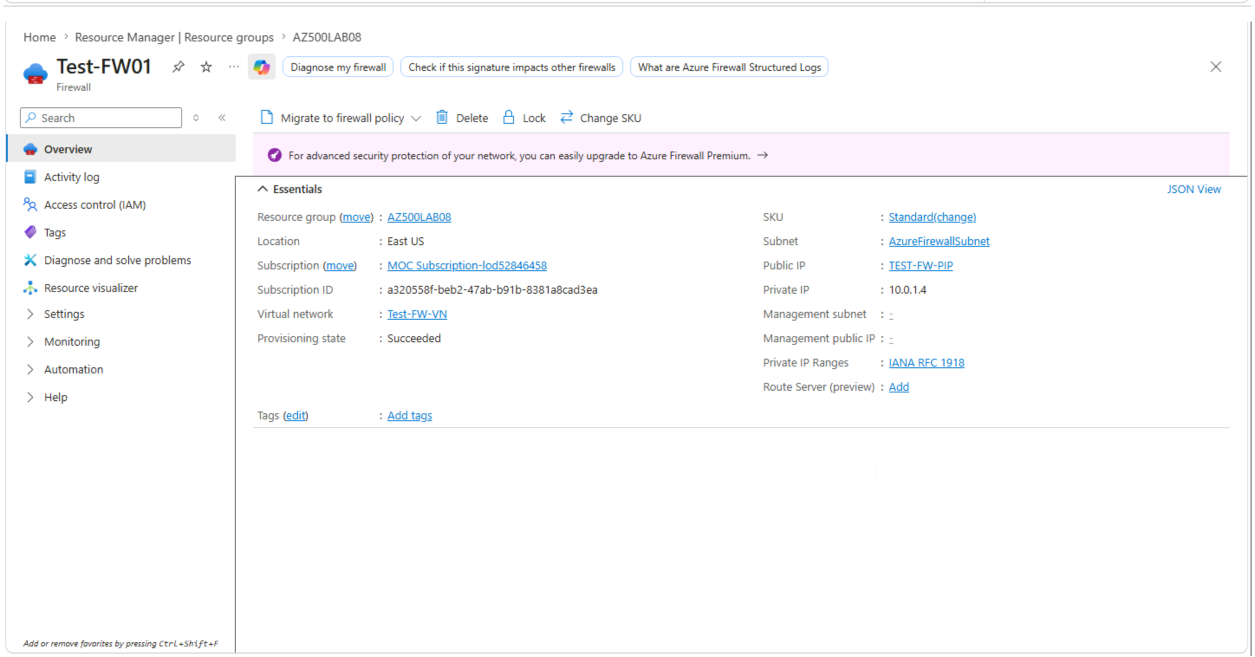
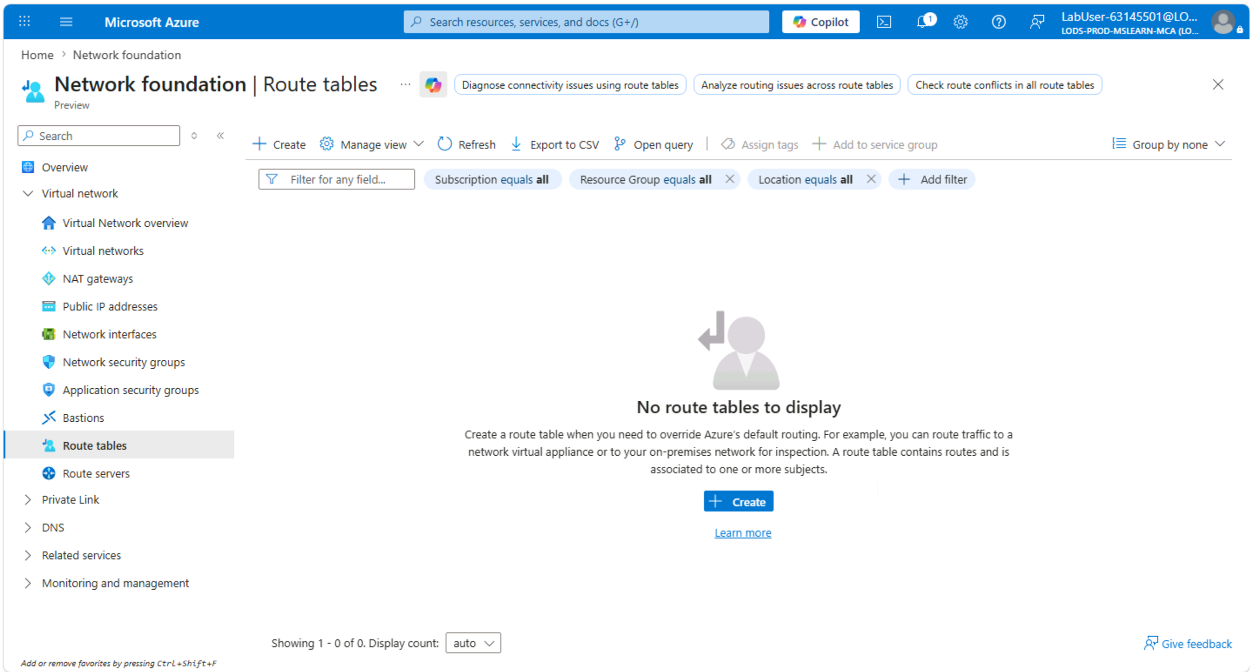
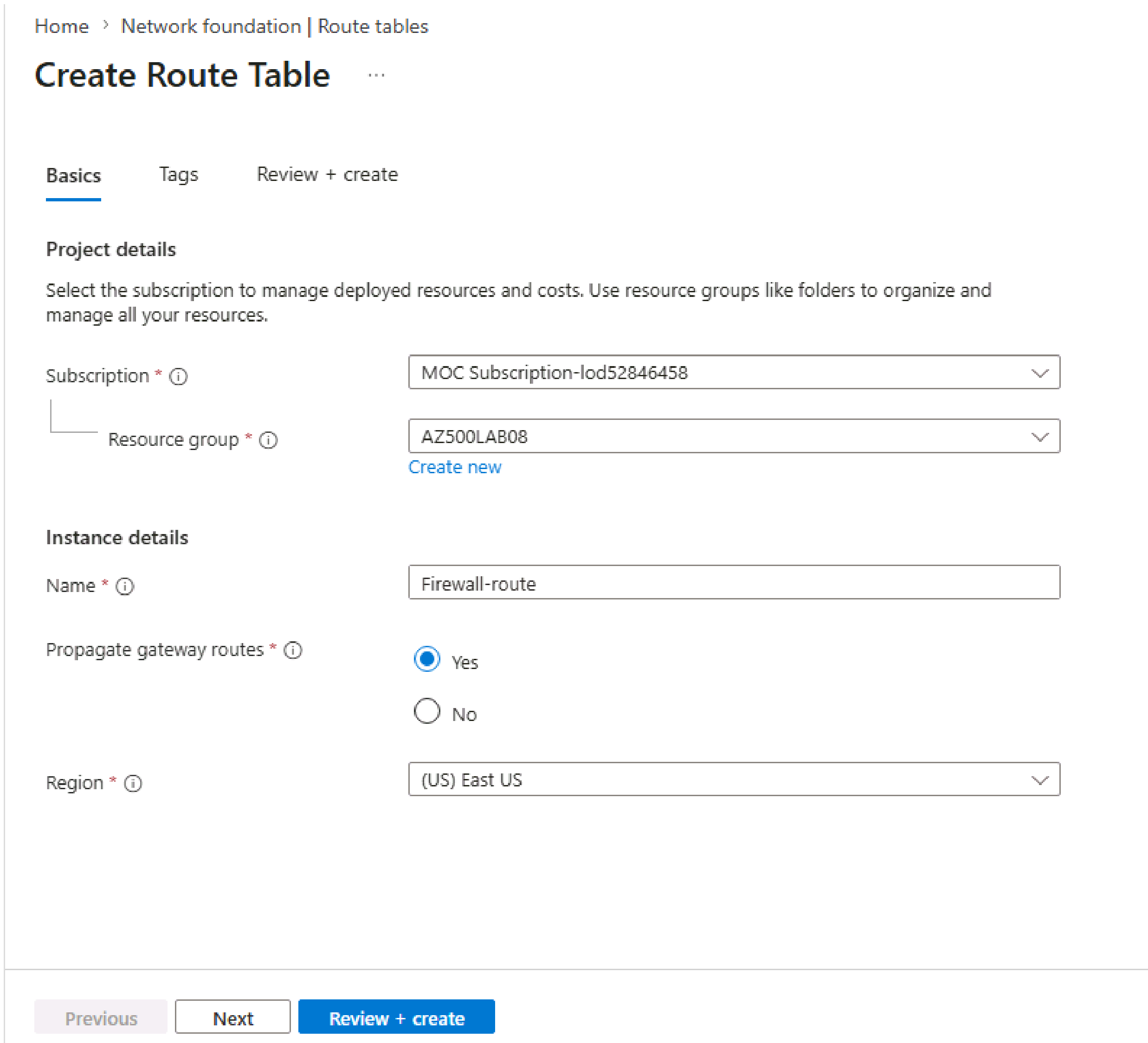
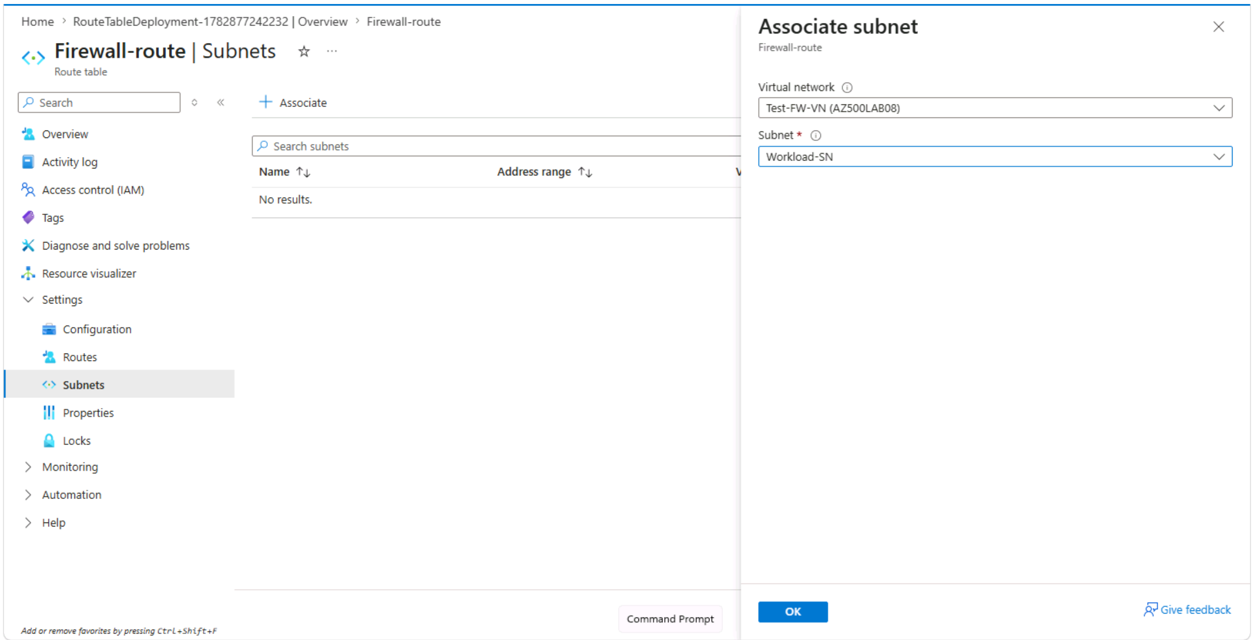
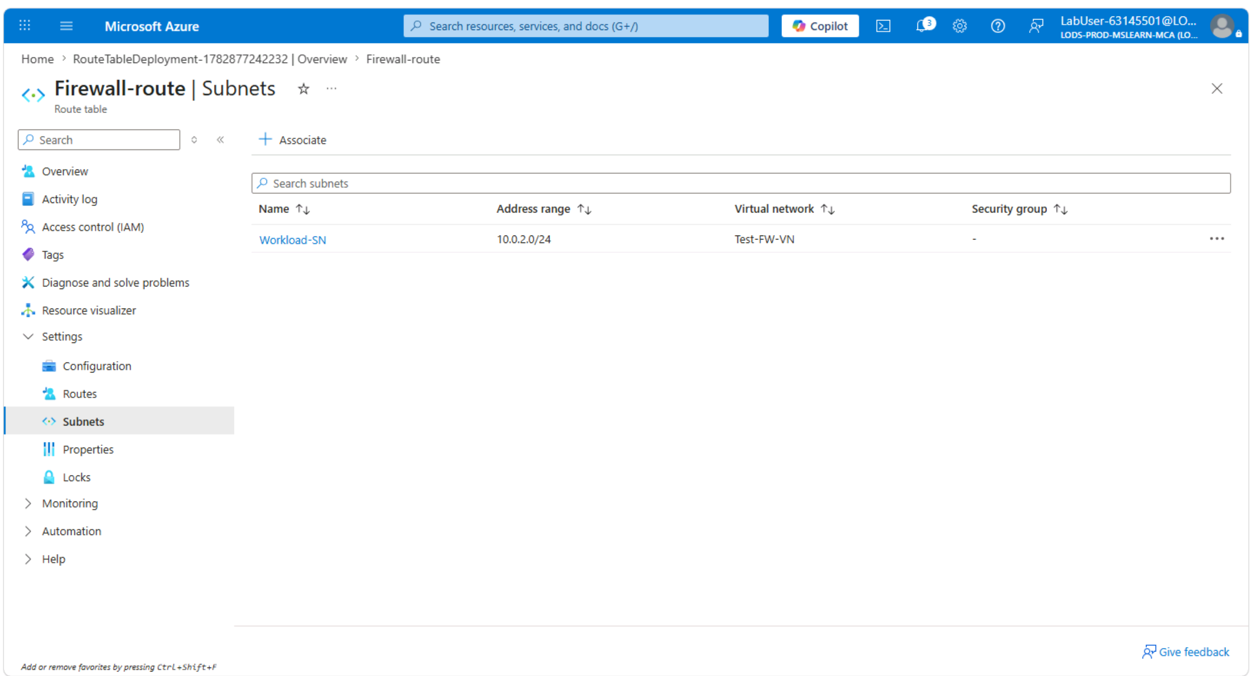
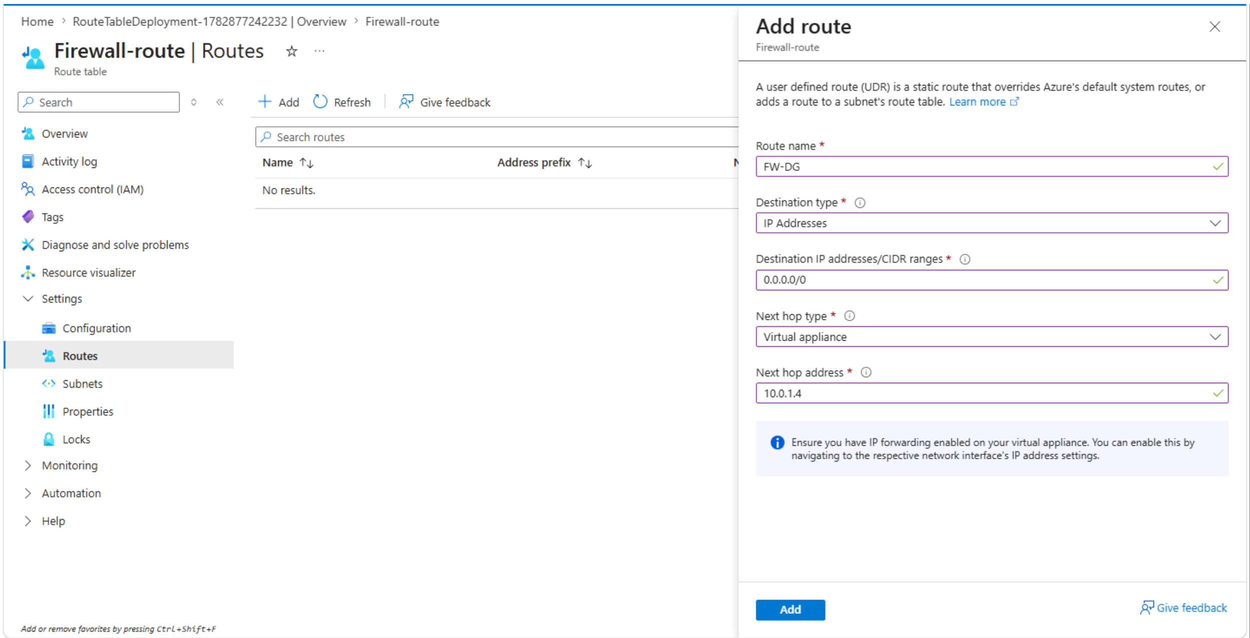

Home > Resource Manager | Resource groups > AZ500LAB08
Test-FW01
Firewall
Diagnose my firewall
Check if this signature impacts other firewalls
What are Azure Firewall Structured Logs
Search
Migrate to firewall policy
Delete
Lock
Change SKU
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Monitoring
Automation
Help
For advanced security protection of your network, you can easily upgrade to Azure Firewall Premium. →
Essentials
JSON View
Resource group (move) : AZ500LAB08
Location : East US
Subscription (move) : MOC Subscription-lod52846458
Subscription ID : a320558f-beb2-47ab-b91b-8381a8cad3ea
Virtual network : Test-FW-VN
Provisioning state : Succeeded
SKU : Standard(change)
Subnet : AzureFirewallSubnet
Public IP : TEST-FW-PIP
Private IP : 10.0.1.4
Management subnet : -
Management public IP : -
Private IP Ranges : IANA RFC 1918
Route Server (preview) : Add
Tags (edit) : Add tags
Add or remove favorites by pressing Ctrl+Shift+F

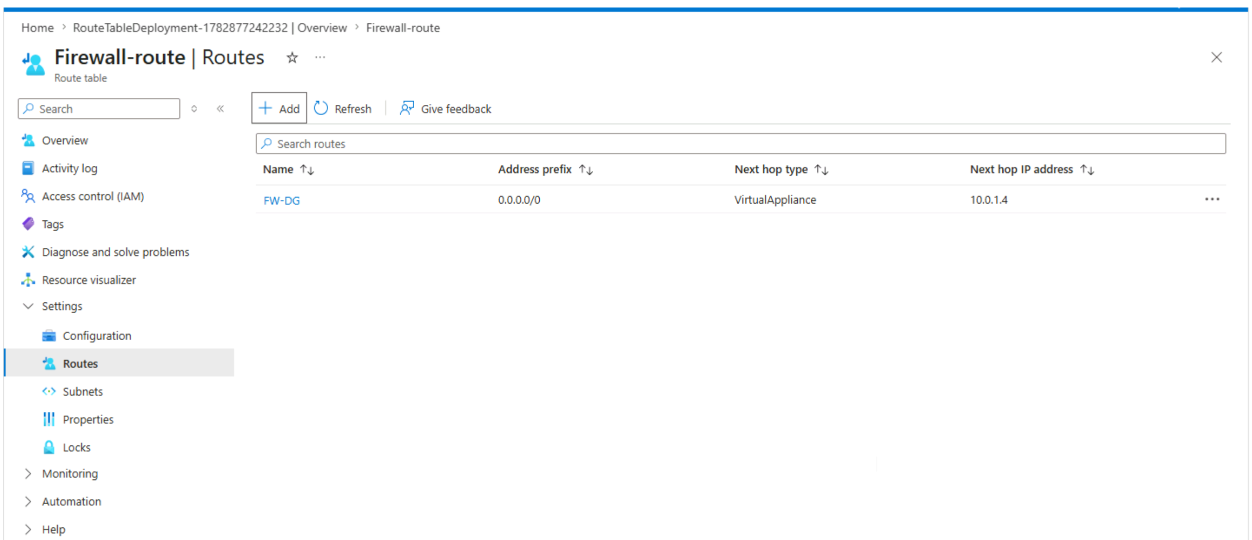

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
LODS-PROD-MSLEARN-MCA (LO...
Home > Network foundation
Network foundation | Route tables
Preview
Diagnose connectivity issues using route tables
Analyze routing issues across route tables
Check route conflicts in all route tables
Search
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Related services
Monitoring and management
Add or remove favorites by pressing Ctrl+Shift+F
Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
No route tables to display
Create a route table when you need to override Azure's default routing. For example, you can route traffic to a network virtual appliance or to your on-premises network for inspection. A route table contains routes and is associated to one or more subjects.
Create
Learn more
Showing 1 - 0 of 0. Display count:
auto
Give feedback

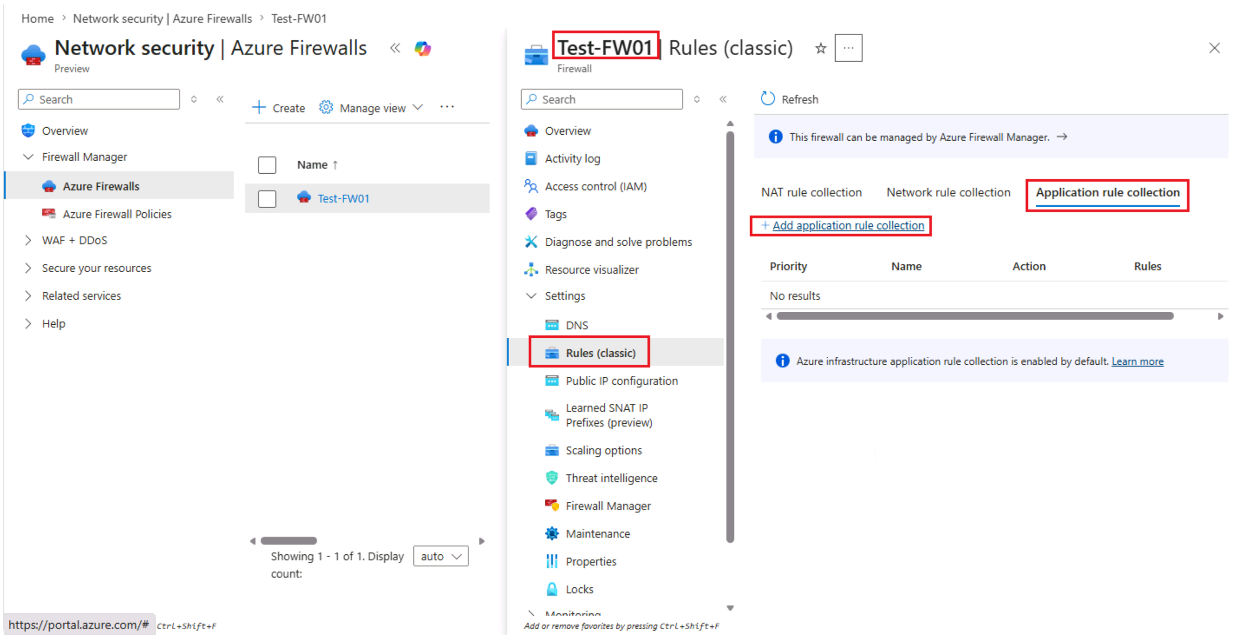

Home > Network foundation | Route tables
Create Route Table ...
Basics
Tags
Review + create
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
MOC Subscription-lod52846458
Resource group *
AZ500LAB08
Create new
Instance details
Name *
Firewall-route
Propagate gateway routes *
Yes
No
Region *
(US) East US
Previous
Next
Review + create

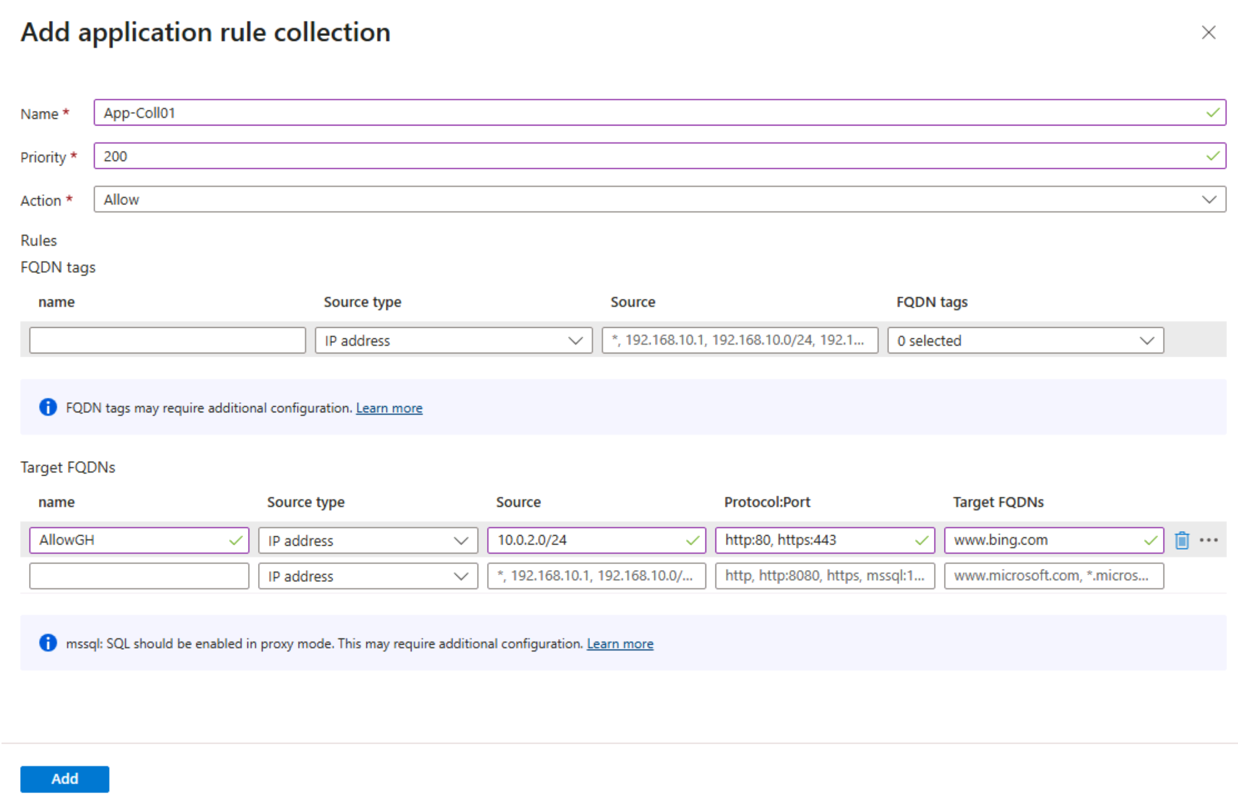

Home > RouteTableDeployment-1782877242232 | Overview > Firewall-route
Firewall-route | Subnets
Route table
Search
Associate
Search subnets
Name
Address range
No results.
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Command Prompt
Associate subnet
Firewall-route
Virtual network
Test-FW-VN (AZ500LAB08)
Subnet *
Workload-SN
OK
Give feedback

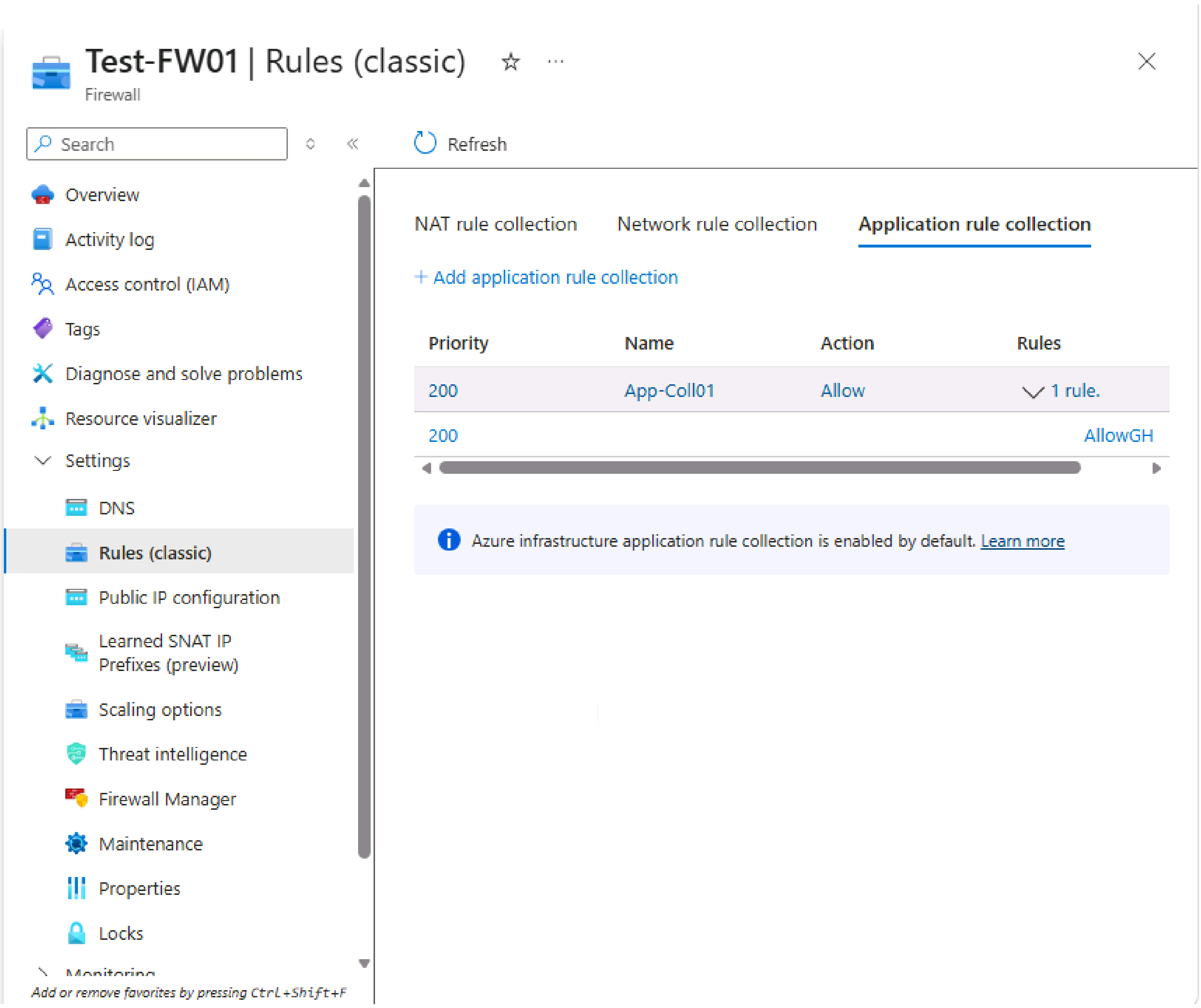

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63145501@LO...
Home > RouteTableDeployment-1782877242232 | Overview > Firewall-route
Firewall-route | Subnets
Route table
Associate
Search subnets
Name
Address range
Virtual network
Security group
Workload-SN
10.0.2.0/24
Test-FW-VN
-
Give feedback

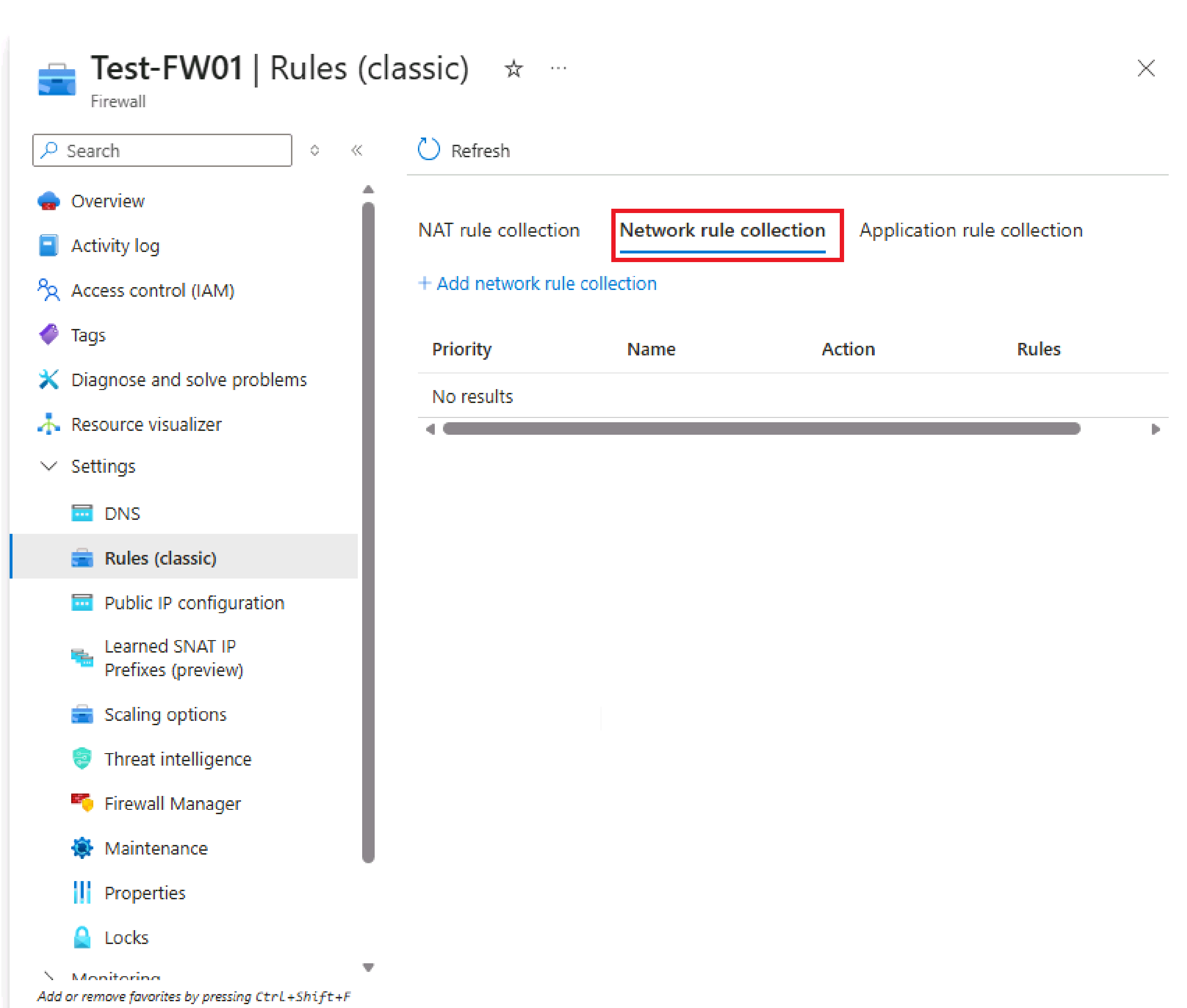

Home > RouteTableDeployment-1782877242232 | Overview > Firewall-route
Firewall-route | Routes
Route table
Search
Add
Refresh
Give feedback
Search routes
Name
Address prefix
No results.
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Add or remove favorites by pressing Ctrl+Shift+F
Add route
Firewall-route
A user defined route (UDR) is a static route that overrides Azure's default system routes, or adds a route to a subnet's route table. Learn more
Route name *
FW-DG
Destination type *
IP Addresses
Destination IP addresses/CIDR ranges *
0.0.0.0/0
Next hop type *
Virtual appliance
Next hop address *
10.0.1.4
Ensure you have IP forwarding enabled on your virtual appliance. You can enable this by navigating to the respective network interface's IP address settings.
Add
Give feedback

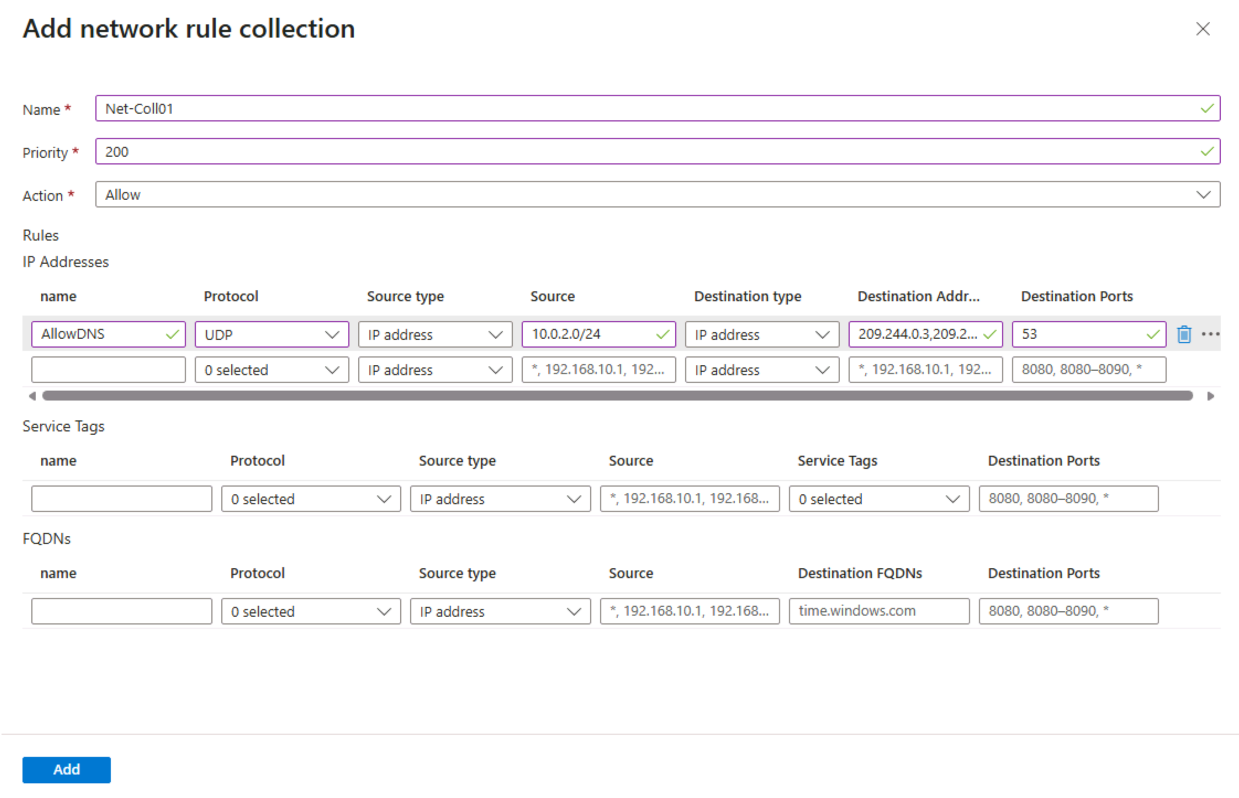

Home > RouteTableDeployment-1782877242232 | Overview > Firewall-route
Firewall-route | Routes
Route table
Search
Add
Refresh
Give feedback
Search routes
Name
Address prefix
Next hop type
Next hop IP address
FW-DG
0.0.0.0/0
VirtualAppliance
10.0.1.4
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help

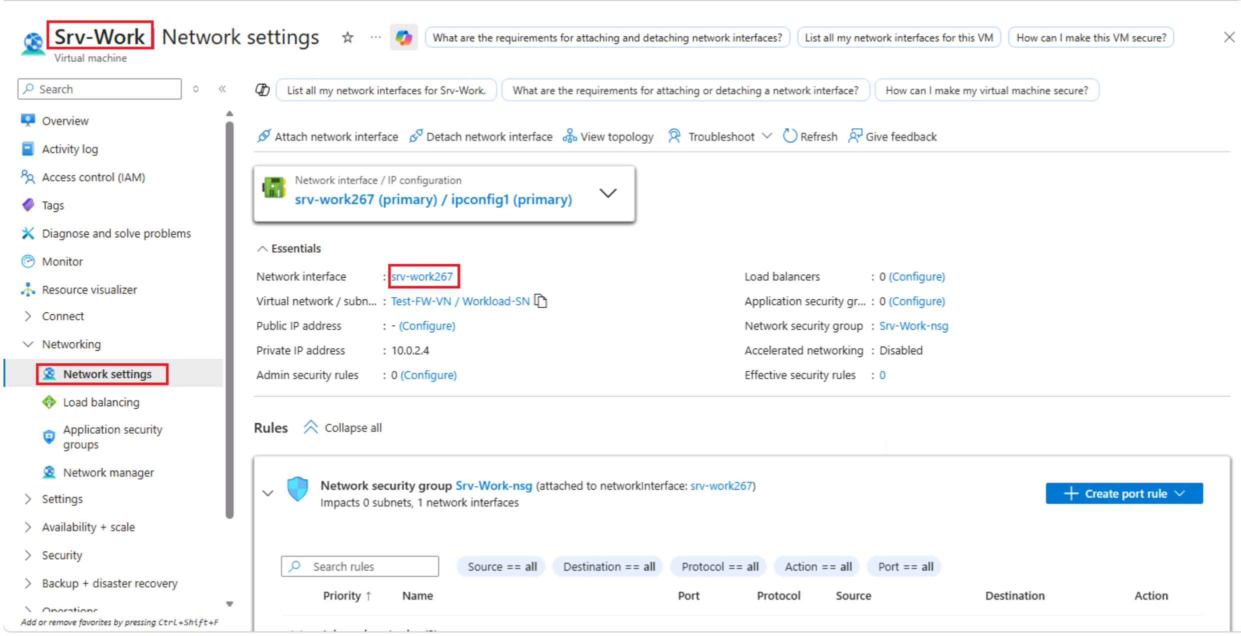

Home > Network security | Azure Firewalls > Test-FW01
Network security | Azure Firewalls
Preview
Search
Overview
Firewall Manager
Azure Firewalls
Azure Firewall Policies
WAF + DDoS
Secure your resources
Related services
Help
Create
Manage view
Name ↑
Test-FW01
Showing 1 - 1 of 1. Display count:
auto
https://portal.azure.com/#
Test-FW01 | Rules (classic)
Firewall
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
DNS
Rules (classic)
Public IP configuration
Learned SNAT IP Prefixes (preview)
Scaling options
Threat intelligence
Firewall Manager
Maintenance
Properties
Locks
Monitoring
Add or remove favorites by pressing Ctrl+Shift+F
Refresh
This firewall can be managed by Azure Firewall Manager. →
NAT rule collection
Network rule collection
Application rule collection
+ Add application rule collection
Priority
Name
Action
Rules
No results
Azure infrastructure application rule collection is enabled by default. Learn more

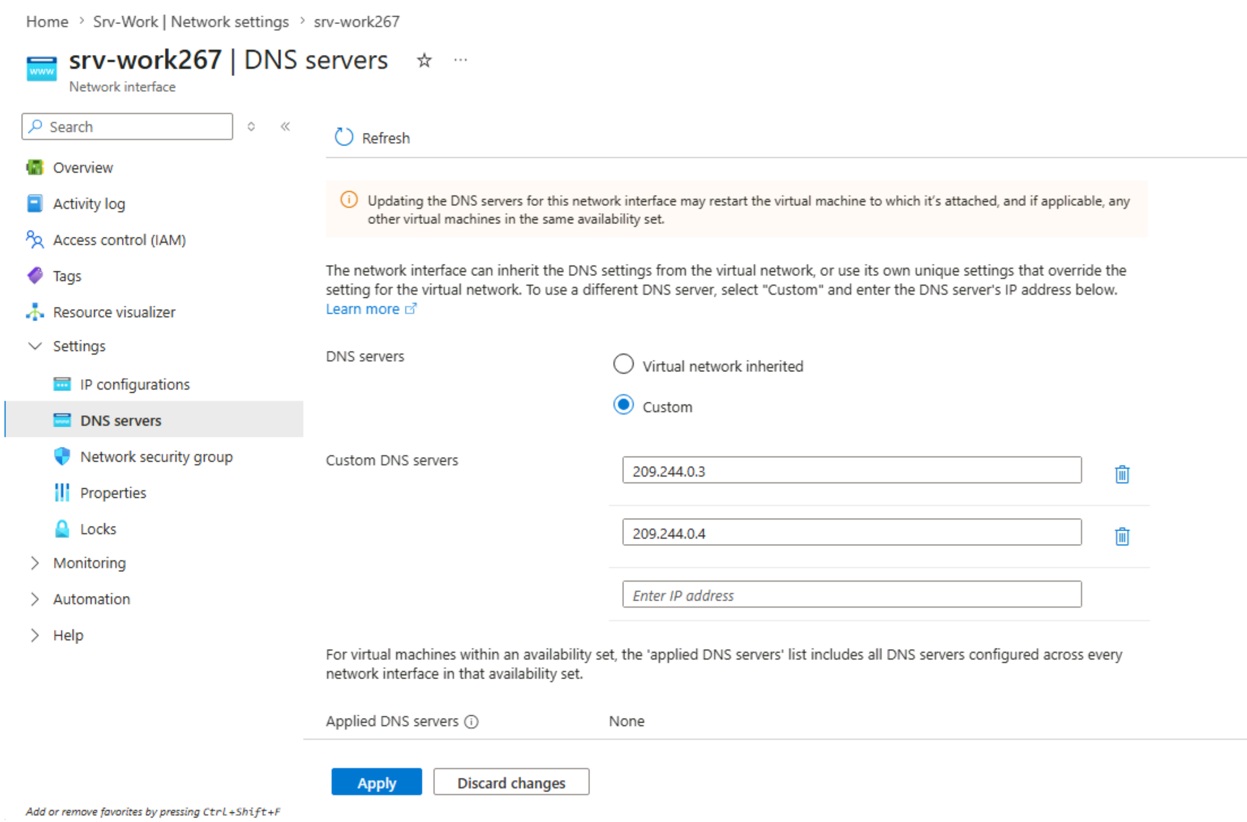

Add application rule collection
Name *
App-Coll01
Priority *
200
Action *
Allow
Rules
FQDN tags
name
Source type
Source
FQDN tags
IP address
*, 192.168.10.1, 192.168.10.0/24, 192.1...
0 selected
FQDN tags may require additional configuration. Learn more
Target FQDNs
name
Source type
Source
Protocol:Port
Target FQDNs
AllowGH
IP address
10.0.2.0/24
http:80, https:443
www.bing.com
IP address
*, 192.168.10.1, 192.168.10.0/...
http, http:8080, https, mssql:1...
www.microsoft.com, *.micros...
mssql: SQL should be enabled in proxy mode. This may require additional configuration. Learn more
Add

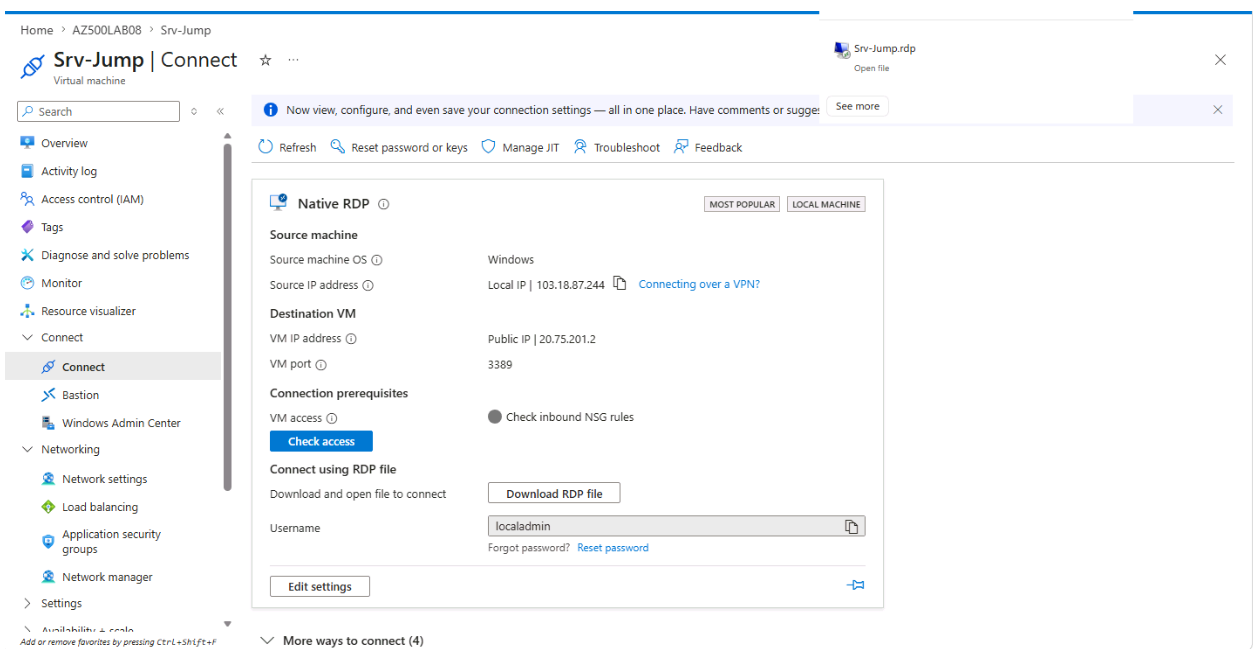

Test-FW01 | Rules (classic)
Firewall
Search
Refresh
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
DNS
Rules (classic)
Public IP configuration
Learned SNAT IP Prefixes (preview)
Scaling options
Threat intelligence
Firewall Manager
Maintenance
Properties
Locks
Add or remove favorites by pressing Ctrl+Shift+F
NAT rule collection
Network rule collection
Application rule collection
+ Add application rule collection
Priority
Name
Action
Rules
200
App-Coll01
Allow
1 rule.
200
AllowGH
Azure infrastructure application rule collection is enabled by default. Learn more

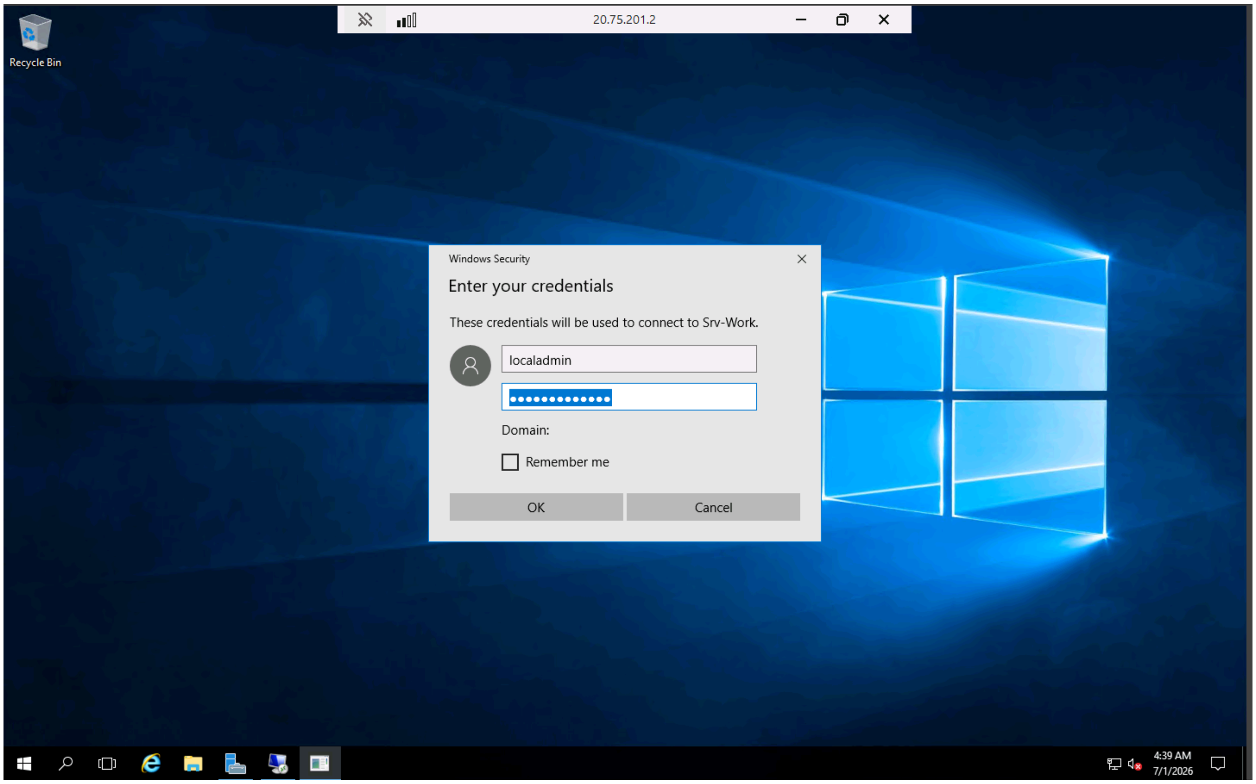

Test-FW01 | Rules (classic)
Firewall
Search
Refresh
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
DNS
Rules (classic)
Public IP configuration
Learned SNAT IP Prefixes (preview)
Scaling options
Threat intelligence
Firewall Manager
Maintenance
Properties
Locks
Add or remove favorites by pressing Ctrl+Shift+F
NAT rule collection
Network rule collection
Application rule collection
+ Add network rule collection
Priority
Name
Action
Rules
No results

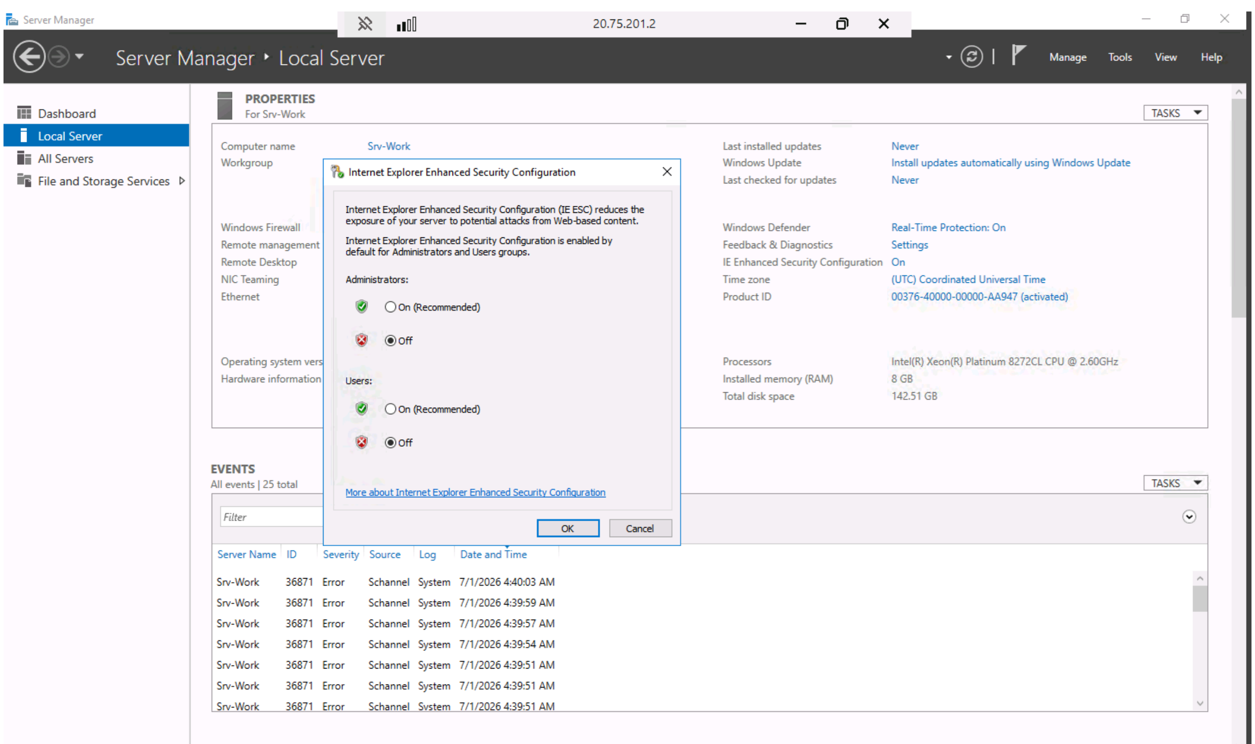

Add network rule collection
Name *
Net-Coll01
Priority *
200
Action *
Allow
Rules
IP Addresses
name
Protocol
Source type
Source
Destination type
Destination Addr...
Destination Ports
AllowDNS
UDP
IP address
10.0.2.0/24
IP address
209.244.0.3,209.2...
53
0 selected
IP address
*, 192.168.10.1, 192...
IP address
*, 192.168.10.1, 192...
8080, 8080–8090, *
Service Tags
name
Protocol
Source type
Source
Service Tags
Destination Ports
0 selected
IP address
*, 192.168.10.1, 192.168...
0 selected
8080, 8080–8090, *
FQDNs
name
Protocol
Source type
Source
Destination FQDNs
Destination Ports
0 selected
IP address
*, 192.168.10.1, 192.168...
time.windows.com
8080, 8080–8090, *
Add

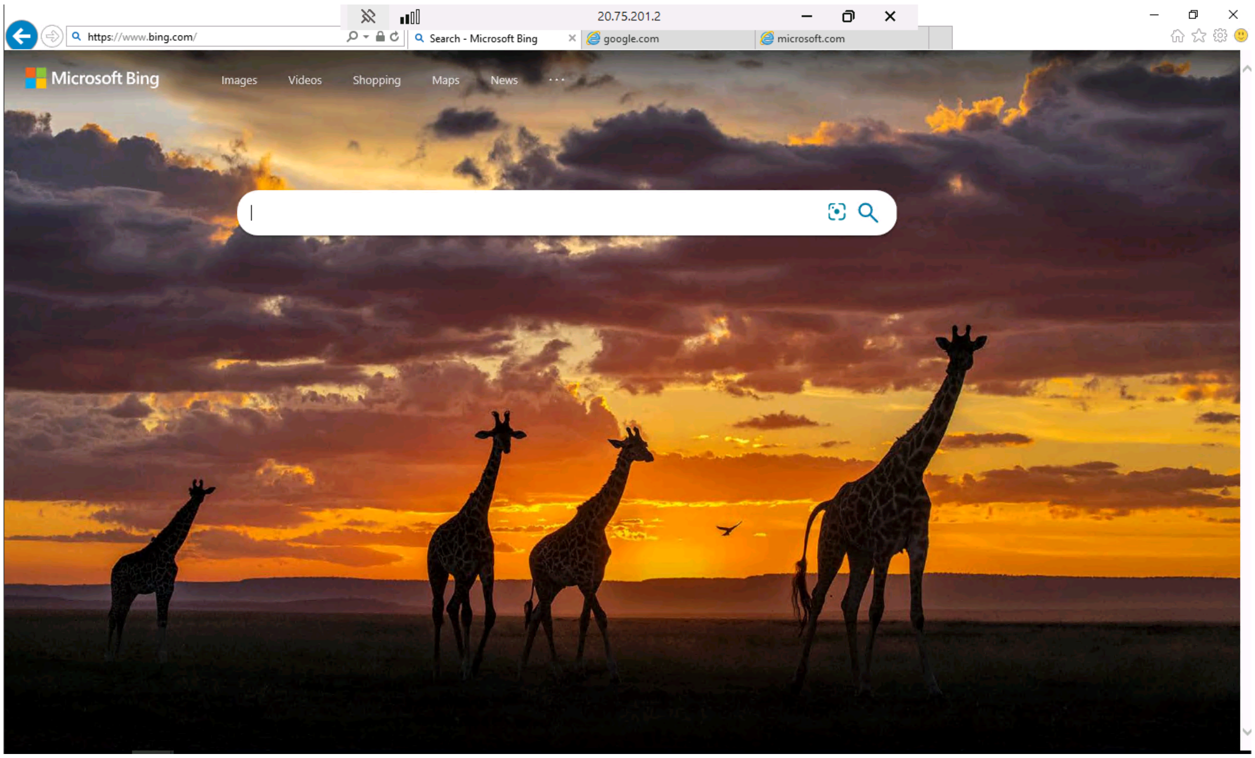

Srv-Work
Network settings
Virtual machine
What are the requirements for attaching and detaching network interfaces?
List all my network interfaces for this VM
How can I make this VM secure?
Search
List all my network interfaces for Srv-Work.
What are the requirements for attaching or detaching a network interface?
How can I make my virtual machine secure?
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings
Availability + scale
Security
Backup + disaster recovery
Add or remove favorites by pressing Ctrl+Shift+F
Attach network interface
Detach network interface
View topology
Troubleshoot
Refresh
Give feedback
Network interface / IP configuration
srv-work267 (primary) / ipconfig1 (primary)
Essentials
Network interface : srv-work267
Virtual network / subn... : Test-FW-VN / Workload-SN
Public IP address : - (Configure)
Private IP address : 10.0.2.4
Admin security rules : 0 (Configure)
Load balancers : 0 (Configure)
Application security gr... : 0 (Configure)
Network security group : Srv-Work-nsg
Accelerated networking : Disabled
Effective security rules : 0
Rules
Collapse all
Network security group Srv-Work-nsg (attached to networkInterface: srv-work267)
Impacts 0 subnets, 1 network interfaces
Create port rule
Search rules
Source == all
Destination == all
Protocol == all
Action == all
Port == all
Priority ↑
Name
Port
Protocol
Source
Destination
Action

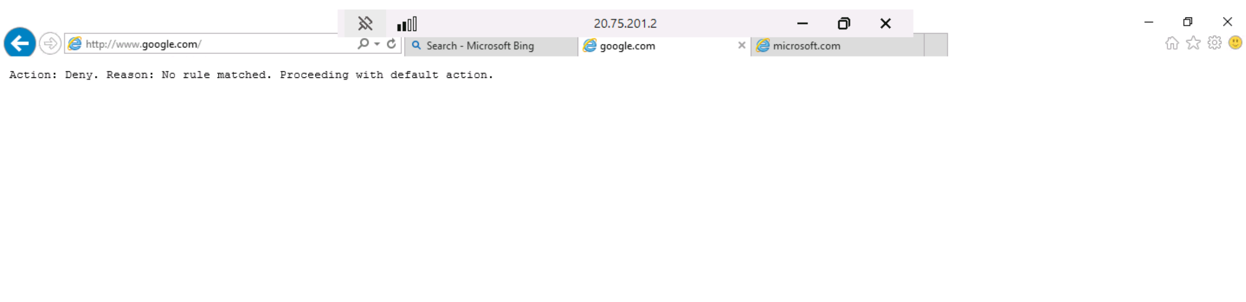

Home > Srv-Work | Network settings > srv-work267
srv-work267 | DNS servers
Network interface
Search
Refresh
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Settings
IP configurations
DNS servers
Network security group
Properties
Locks
Monitoring
Automation
Help
Updating the DNS servers for this network interface may restart the virtual machine to which it's attached, and if applicable, any other virtual machines in the same availability set.
The network interface can inherit the DNS settings from the virtual network, or use its own unique settings that override the setting for the virtual network. To use a different DNS server, select "Custom" and enter the DNS server's IP address below.
Learn more
DNS servers
Virtual network inherited
Custom
Custom DNS servers
209.244.0.3
209.244.0.4
Enter IP address
For virtual machines within an availability set, the 'applied DNS servers' list includes all DNS servers configured across every network interface in that availability set.
Applied DNS servers
None
Apply
Discard changes
Add or remove favorites by pressing Ctrl+Shift+F

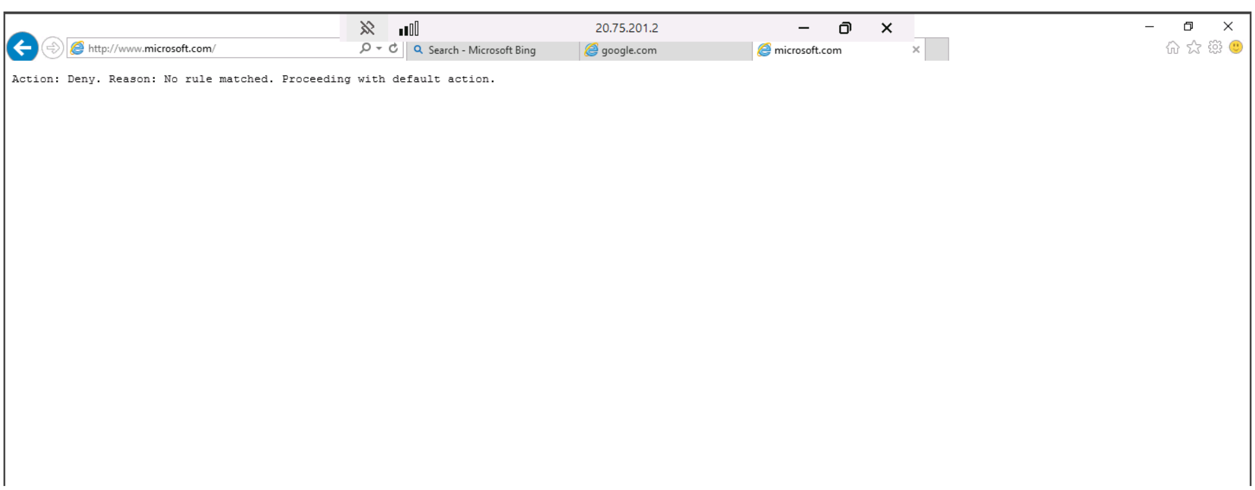

Home > AZ500LAB08 > Srv-Jump
Srv-Jump | Connect
Virtual machine
Srv-Jump.rdp
Open file
Search
Now view, configure, and even save your connection settings — all in one place. Have comments or sugges
See more
Refresh
Reset password or keys
Manage JIT
Troubleshoot
Feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings
Native RDP
MOST POPULAR
LOCAL MACHINE
Source machine
Source machine OS
Windows
Source IP address
Local IP | 103.18.87.244
Connecting over a VPN?
Destination VM
VM IP address
Public IP | 20.75.201.2
VM port
3389
Connection prerequisites
VM access
Check inbound NSG rules
Check access
Connect using RDP file
Download and open file to connect
Download RDP file
Username
localadmin
Forgot password? Reset password
Edit settings
More ways to connect (4)
Add or remove favorites by pressing Ctrl+Shift+F


Recycle Bin
20.75.201.2
Windows Security
Enter your credentials
These credentials will be used to connect to Srv-Work.
localadmin
Domain:
Remember me
OK
Cancel
4:39 AM
7/1/2026


Server Manager
20.75.201.2
Server Manager • Local Server
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage Services
PROPERTIES
For Srv-Work
TASKS
Computer name
Srv-Work
Workgroup
Last installed updates
Never
Windows Update
Install updates automatically using Windows Update
Last checked for updates
Never
Windows Firewall
Remote management
Remote Desktop
NIC Teaming
Ethernet
Windows Defender
Real-Time Protection: On
Feedback & Diagnostics
Settings
IE Enhanced Security Configuration
On
Time zone
(UTC) Coordinated Universal Time
Product ID
00376-40000-00000-AA947 (activated)
Operating system vers
Hardware information
Processors
Intel(R) Xeon(R) Platinum 8272CL CPU @ 2.60GHz
Installed memory (RAM)
8 GB
Total disk space
142.51 GB
Internet Explorer Enhanced Security Configuration
Internet Explorer Enhanced Security Configuration (IE ESC) reduces the exposure of your server to potential attacks from Web-based content.
Internet Explorer Enhanced Security Configuration is enabled by default for Administrators and Users groups.
Administrators:
On (Recommended)
Off
Users:
On (Recommended)
Off
More about Internet Explorer Enhanced Security Configuration
OK
Cancel
EVENTS
All events | 25 total
Filter
Server Name
ID
Severity
Source
Log
Date and Time
Srv-Work 36871 Error Schannel System 7/1/2026 4:40:03 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:59 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:57 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:54 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:51 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:51 AM
Srv-Work 36871 Error Schannel System 7/1/2026 4:39:51 AM


20.75.201.2
https://www.bing.com/
Search - Microsoft Bing
google.com
microsoft.com
Microsoft Bing
Images
Videos
Shopping
Maps
News


20.75.201.2
http://www.google.com/
Search - Microsoft Bing
google.com
microsoft.com
Action: Deny. Reason: No rule matched. Proceeding with default action.


20.75.201.2
http://www.microsoft.com/
Search - Microsoft Bing
google.com
microsoft.com
Action: Deny. Reason: No rule matched. Proceeding with default action.